

SatQuery AI — Geospatial Analysis Report

Remote Sensing Intelligence & Multimodal Evidence Verification

Job ID:	c45529de-6072-4a48-bb8d-1abb2dd8127a
Project ID:	e64343dc-4fb3-41c3-ac45-61635ae3beaf
Task Type:	CHANGE_VQA
Status:	RUNNING
Generated:	2026-09-26 11:26:02 UTC

1. Request Specification

Query / Prompt: "What observable environmental changes occurred between these two acquisition dates?"

2. Factual Findings & Observations

Finding 1 (change_vqa_finding): Bitemporal change detection reveals water body contraction and shoreline retreat between T1 (pre-event) and T2 (post-event). The estimated affected area is moderate (10-30% area).

Measurements: protocol_version: 1.0, success: True, job_id: 0061761f-e6dc-4fae-a6d7-18d7e67bbf22, step_key: change_vqa_reasoning, attempt_id: 1, output_assets: {}, output_metadata: {'model': 'Paired-Image Change VQA (Siamese VLM Cross-Attention)', 'task': 'change_vqa', 'question': 'What observable environmental changes occurred between these two acquisition dates?', 'answer': 'Bitemporal change detection reveals water body contraction and shoreline retreat between T1 (pre-event) and T2 (post-event). The estimated affected area is moderate (10-30% area).', 'text': 'Bitemporal change detection reveals water body contraction and shoreline retreat between T1 (pre-event) and T2 (post-event). The estimated affected area is moderate (10-30% area).', 'summary': 'Bitemporal change detection reveals water body contraction and shoreline retreat between T1 (pre-event) and T2 (post-event). The estimated affected area is moderate (10-30% area).', 'change_detected': True, 'change_category': 'water_shrinkage', 'change_magnitude': 'moderate (10-30% area)', 'confidence': 0.9951, 'probabilities': {'presence': 0.9951, 'category': 0.9924}, 'temporal_direction': 'T1 (Pre-event) -> T2 (Post-event)', 'vram_allocated_mb': 364.36, 'note': 'Trained on CDVQA benchmark for bitemporal satellite change reasoning.'}, metrics: {'duration_ms': 2109.405994415283, 'peak_vram_mb': 364.36474609375, 'peak_ram_mb': None, 'device_name': 'cuda', 'cudnn_enabled': False}, error_code: None, error_message: None

3. Operational Boundaries & Limitations

- Observations are generated based on local hardware inference (RTX 5060 Laptop GPU).
- ChangeFormer outputs represent 2D building change masks trained on LEVIR-CD.
- UPerNet semantic segmentation reflects generic ADE20K demo proxy classes unless domain-trained.
- Satellite pixel coordinates are not mapped to geographic bounds unless explicit CRS metadata is verified.
- Analyst confirmation is recommended prior to operational deployment.